An Unsupervised Approach to Indirect Observation of Dark Matter in Rotation Curves

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Introduction

Dark matter is one of the longest-enduring mysteries of astrophysics. Although theorized due to observed differences in rotational velocities of spiral galaxies and the mathematical predictions derived from the composition of visible mass in a galaxy, science still has no complete explanation for whether this mass is really even the correct explanation for the calculated anomalies in rotation curves. According to Newtonian mechanics, the velocity of matter on the outside rings of a galaxy should decrease as the radius increases. However, empirical data find that these rotational curves are actually flat, meaning that this outside matter moves much faster than calculated (Rubin & Ford, 1970). This finding suggests that the best possible explanation is that of another mass which cannot be observed, but has a gravitational effect on the rotation curves of the galaxies it inhabits. Although studies have tried to empirically prove the existence of dark matter, none have yet succeeded (Akerib, Daniel et al, 2017).

Machine learning is defined by Internal Business Machines Corporation as “the subset of artificial intelligence (AI) focused on algorithms that can ‘learn’ the patterns of training data and, subsequently, make accurate inferences about new data” (Bergmann, Dave, 2025). Machine learning offers an approach to detecting patterns in these galaxies that suggest and support the correct conditions for the possibility (and assumption) of dark matter existing. By inputting rotational data from galaxies, such as rotational velocity, a machine learning algorithm should be able to predict what the mass composition of the galaxy is and compare it to others. This study uses an unsupervised algorithm, meaning that the algorithm will only be trained on unlabeled data and expect it to derive similarities and organize them on its own. This was the best choice for this study, as an unsupervised algorithm can detect patterns in data that a researcher may overlook at first glance. In this way, I’ll be able to use an algorithm that can detect subtle, but important, similarities in galaxies that may be determining factors of mass composition and amount of dark matter. In addition, I’m addressing a significant gap in research; most studies that directly address mass composition and dark matter detection use supervised algorithms and train them on labeled data of different mass compositions (Wu, Sirui, et al, 2024). In addition, most studies begin with the assumption that their algorithms are accurate; few include measurements of accuracy like mean silhouette score or mean ARI. Therefore, the goal of this study is to determine whether unsupervised machine learning algorithms can accurately identify patterns in spiral galaxy rotation curve anomalies that suggest the presence of dark matter compared to Baryonic mass predictions.

Foundation

The studies that are foundational to this research are also the foundational studies for the hypothesis of the existence of dark matter. Since the first systematic spectroscopic (spectroscopic meaning analyzing how light interacts with matter to determine physical properties) measurements of external galaxies have shown unexpectedly flat outer rotation curves of spiral galaxies (Rubin & Ford), and observed orbital velocities at large radii remain high compared with what is predicted by baryonic mass, defined as empirically and directly observed mass, it provides the foundation for further inquiry into the indirect mathematical observation of the source of this mysterious gravitational force acting upon the outer rotation curves of galaxies. As a response to these observations and mathematical anomalies, literature

has provided two main classifications of explanations: firstly, physical explanations, such as dark matter haloes, an alternative gravitational model, or changes in dark-matter microphysics, such as weakly-interacting massive particles (WIMPs) versus sterile neutrinos, versus primordial black holes. Secondly, improvements made to observational techniques that either allow for an observation of dark matter or an observation of rotation curves that do in fact align with gravitational theory according to baryonic mass. In direct parallel to the development of dark matter theory, machine learning is becoming ever more prominent in today's commercial and scientific societies. This review of the current literature in both of these interacting fields aims to provide a clear methodology based on all data available to ensure the accuracy of research and to establish a clear methodological approach as well as a baseline expectation for the results of this study and its limitations.

Literature Review

1 - Baryonic-Dark Relationship

A large amount of literature is observational research. This research is imperative to this study as it defines the empirical baseline against which any machine learning algorithm must be judged for accuracy. These reviews of rotation curves compile classifications of rotation-curve morphologies, meaning visual classification, and document measurement uncertainties. Studies that take and record this observational data, such as in the SPARC database (Lelli et al.), which is crucial for this study, and the DISKMASS survey (Martinsson et al.), provide these calibrated rotational curves and baryonic decompositions that are necessary for algorithmic analysis. More specifically, the radial acceleration relation present in the SPARC database (Lelli et al.) further narrows the study by providing an empirical relationship between baryonic acceleration and total observed acceleration across galaxies. Thus, this prior research evidently provides a baseline as to where the greater the residual from this observed relationship, the larger the apparent anomaly.

Although there are guidelines and constraints for this research, it also shows clear diversity in the holistic behavior of rotational curves. In De Blok's discussion of the core-cusp problem, it's evident that dark-matter density profiles are not in any way uniform. In fact, low-surface-brightness and dwarf galaxies show core-like matter profiles, directly contradicting collisionless N-body predictions, suggesting the need for different or more complex physics to explain this phenomenon. In addition, works on self-interacting dark matter (Kamada et al.) show that alternate explanations for dark matter physics can recreate similar rotation-curve morphologies, while further studies (Bacon) emphasize more complex models, even going so far as to suggest changes to gravitational theory as an alternative to dark matter, which is also clearly supported by a lack of direct observational data as opposed to simulations and mathematical observations.

In addition, studies and meta-analyses that examine populations of galaxies with low dark matter fractions or galaxies at high redshifts, meaning further away, that contain flat curves imply the existence of relevant outlier cases, which must be taken into account when inputting data into an algorithm. For example, Drew et al. used spectroscopic data from the KMOS instrument to find that galaxies at high redshifts do not represent outliers in the baryonic

fraction. However, these individualized galaxies vary massively, presenting a high probability of selecting an ‘outlier’ galaxy when inputting data into algorithms (Drew et al.). These results were independently confirmed by Sharma et al., more recently, using a separate, less direct methodology (cosmological hydrodynamic simulations) (Sharma et al.). Finally, Turner et al. link low and medium-redshift galaxies using photometric and spectroscopic data to find evolutionary patterns. In doing so, Turner et al. independently confirmed that redshift does not play a role in the baryonic fraction.

2 - Machine Learning in Astrophysics

Although machine learning has been applied in various ways to the field of astrophysics, this paper aims to address a significant and glaring methodological gap in research — although studies using unsupervised machine learning algorithms for dark matter anomaly analysis do exist, they seldom begin with testing the accuracy of the machine learning algorithm before implementing it for analysis. Every study listed below that uses unsupervised machine learning algorithms does not have a pre-analysis accuracy measurement. In turn, this study will address the accuracy of machine learning in doing so.

Machine learning has been applied to many astronomical tasks. For example, supervised approaches have been applied to estimate masses or classify morphological data when it is available and labeled (Ntampaka et al.). The success of supervised machine learning makes it apparent that machine learning is capable of extracting meaningful patterns from morphological and kinematic data, although the data is mostly limited to labeled training sets. This presents a clear limitation, however, to these studies as it constrains the algorithm’s capabilities to discover physically meaningful clusters of rotation-curve anomalies.

Therefore, literature surrounding the use of supervised and unsupervised algorithms is especially relevant to this research. Hocking et al. and Cheng et al. both show that clustering algorithms can create taxonomies of galaxy morphologies without labeled data; therefore, although not entirely relevant to the goal of this study, they prove that unsupervised algorithms do have value in the field. Mohale & Lochner show that self-supervised representation learning, such as BYOL-style pretraining, is useful because it can accurately cluster and detect anomalies in astronomical imaging. Argüelles & Collazo and Garcia-Arroyo et al. specifically use neural networks for rotation-curve fitting and reconstruction of astronomical data — in this sense, neural networks may be useful to reconstruct or model rotation-curves due to their ability to consistently improve the network’s weights to make more accurate predictions.

The works discussed present a credible foundation for a methodological approach to the research presented in this paper. It classifies strands of machine learning as thus used most accurately: firstly, it classifies self-supervised machine learning as best for learning to derive missing information from incomplete kinematic or morphological data, and for simulating. Secondly, it classifies probabilistic clustering as being valuable to identify sub-populations and outliers. Third, it identifies unsupervised anomaly-detection frameworks separate from probabilistic clustering as valuable to flag rotation-curve profiles that deviate from baryonic-based expectations. As this research study focuses entirely on outlier and anomaly

detection, the literature suggests a possible methodology consisting of probabilistic clustering and/or separate anomaly-detection frameworks.

Several alternative methodological approaches have also been analyzed as possible aids to algorithms. Specifically, rotation-invariant CNNs (Convolutional Neural Networks) demonstrate the usefulness of image interpretation resistant to asymmetrical data by using multiple layers to process data, such as convolutional, pooling, and fully connected layers. In addition, the specific approach used by Mohale & Lochner, combining self-supervised learning, Bayesian-Gaussian Mixture Model (BGMM) clustering (which is entirely probabilistic), and an anomaly-detection framework, has been empirically proven as a near-perfect methodology for rotation-curve analysis.

3 - Datasets

Finally, it is necessary to discuss the limitations of datasets that may be present in this study, specifically the SPARC database. For example, selection biases (such as nearby galaxies or well-researched galaxies) and varying measurement techniques (such as long-slit versus integral-field spectroscopy) may produce differences in sample quality that may adversely affect results. Kassin et al. and Lelli et al. show specifically that behavior changes based on radius. In any database, this must be a controlled variable, or, at the very least, present and considered within an algorithm.

In addition, outlier studies show that the standard deviation away from the average dark matter fraction is high and nontrivial for this study (MNRAS). These cases should be used as a test of the sufficiency of any algorithm. In addition, incomplete data and insufficiencies in specific observational data may lead to inconsistent results. Garcia-Arroyo et al.'s neural network reconstruction suggests that there are viable strategies to evade the inconsistencies of this observational data.

Another justification made for this study is that the SPARC database, a signifier for Spitzer Photometry and Accurate Rotation Curves, best matches the aims of this study. There are two justifications for this. Firstly, the database has high-quality HI+Halpha rotation curves, meaning that the gravitational potential is being traced out to a large radius, which is well-suited for an analysis of outer rotation curves since it allows us to analyze galaxies with a larger deviation. Secondly, the database has a diverse range of galaxy types and characteristics, meaning that this research can likely be applied to a diverse range of galaxy types and characteristics instead of being confounded by variables related to type. The SPARC database was chosen instead of similar databases such as the DISKMASS survey due to its focus on rotation curves at large radii and its analysis of radial acceleration across the entire disk, as compared to DISKMASS's focus on decomposition and its focus on smaller radii.

Method

This study uses a quantitative, computational research design to identify patterns within SPARC data using unsupervised machine learning and k-means clustering, to then identify the accuracy of the algorithm through analysis of the created clusters. The methodology has four main parts: firstly, data acquisition and processing. Secondly, data cleaning and preparation.

Third, model training. Fourth, analysis and prediction. All of these procedures were created and executed using a Python pipeline developed specifically for this research study, including scripts for each major part. Python was selected as the programming language of choice for this project due to its powerful ecosystem and availability of packages. For example, NumPy, Pandas, scikit-learn, and Matplotlib are all packages in the Python ecosystem that make it ideal for data processing, machine learning, and especially for creating visualizations.

The first program in this study comes from a custom Python script in the 'prepare_data.py' file. In this file, the program locates all relevant files within the SPARC database and prepares every piece of useful data into a structured table format. Next, raw rotational velocities and baryonic components are converted into numbers that quantify rotational-curve behavior. This includes summary statistics that describe the magnitude and consistency of deviations between observed rotation curves and baryonic predictions. Finally, the output of this program is placed into the file 'features.csv', in which each row is a singular galaxy and each column is a feature used.

The next crucial program is 'train_model.py'. This stage of the study applied unsupervised learning to the cleaned data. Firstly, the program standardizes the features in features.csv using z-score normalization, which ensures that no single variable dominates the clustering due to its scale. Next, PCA (Principal Component Analysis) is applied. This was chosen as opposed to LDA (Linear Discriminant Analysis) because it requires labeled classes, something that would defeat the point of an unsupervised algorithm, whereas PCA does not. PCA reduces dimensionality by reducing the relationship between features to a smaller number of principal components. After this is done, only three principal components are retained, with PC1 representing the overall rotation-curve deviation magnitude, PC2 representing the Baryonic dominance vs Dark-Matter dominance, and PC3 representing the radial structure of deviations, meaning how deviations change with radius (such as smoothly increasing discrepancies vs more irregular discrepancies). Three principal components were used due to their effectiveness in being physically applicable to rotation curves. However, it is important to note that these descriptions of the principal components are approximations since each principal component represents a linear combination of multiple features for the sake of simplicity and feasibility. This is also a limitation of the study — without the capability to individually analyze each and every feature, any patterns found from this data will be found from approximations.

Then, in the same file, clustering is performed using the k-means algorithm. K-means was chosen because it is specifically compatible with PCA and is accurate in clustering data points. The created clusters are shown in Appendix D, organized by PC1 and PC2. This algorithm was chosen in favor of others described in the literature review due to its interpretability and suitability for clustering using PCA, which will allow this research to make connections between differences in clusters and meaningful interpretations of rotation-curve features. These clusters, physically, are groups of galaxies that share similar structure according to the features along the third dimension of PCA. Therefore, physically, clusters are groupings of data points, with each data point representing one galaxy. In order to avoid choosing the number of clusters arbitrarily and risking simplifying data too much, the program evaluates multiple values for k and calculates the silhouette scores for each. Silhouette scores are a metric used to evaluate how well a clustering algorithm has grouped points, with the score being calculated as $s = \frac{b-a}{max(a,b)}$, with 'a' representing the average distance between a point and all

the other points in the same cluster, which shows how tightly grouped the cluster is, and ‘b’ representing the average distance between a point and points in the nearest neighboring cluster, showing how close it is to another cluster. Values closer to +1 indicate the point is well matched and therefore the clustering was likely successful, a value of 0 indicates that the point is likely on a boundary between clusters, therefore the test is inconclusive with that specific point, and -1 means the point is likely in the wrong cluster and the clustering was unsuccessful. After the program calculates the silhouette scores for each value of k (the number of clusters), it then determines which number is best and continues with that value for k. Finally, the cluster model and PCA are saved for future use with similar data, ensuring repeatability.

The final stage of the methodology extends the results of the clustering to readable data and interprets them. A visualization script (visualize.py) creates the three figures in this study used in the results section, and predict_new.py applies the models created above to new data by loading the saved k-means algorithm and PCA to apply it again, ensuring reproducibility of the experiment with the same or new data, since the exact algorithm and scalers are saved.

Next, it is necessary to determine the accuracy of the analysis done by the machine learning program.

Evaluation Metrics

The first metric we’re going to use for evaluation is the PCA Explained Variance. This metric is a preliminary metric of evaluation to ensure that the original features input are captured accurately in the dimensionality reduction. It explains exactly how much of the original variability amongst the original features is captured by each component of the PCA. To clarify, the ideal outcome is that most of the variance is captured by a few components, which would mean that any results stemming from that are accurate to the original data. This is calculated in a few steps:

1. The original data is standardized, meaning z-scored
2. The covariance matrix of each feature is computed
3. Eigenvalues and Eigenvectors are calculated for the covariance matrix
    a. Eigenvectors are the principal components
    b. Eigenvalues are the amount of variance for that component
4. Explained Variance Ratio is the eigenvalue divided by the sum of all the eigenvalues

Before we move on to the other metrics, it is necessary to delve into some of the mathematics involved in the calculation of PCA Explained Variance, although all of this math is done by the algorithm.

Firstly, z-scoring is a way of standardizing values since certain variables may have distinct ranges or values that could exist that others do not, I.E. one variable might range from 0-1, one variable may be able to contain negatives, one variable may range from 0-300, etc. For each value, the z-score is calculated as $z = \frac{x-\mu}{\sigma}$, where x represents the original value, μ represents the mean of that feature, and σ represents the standard deviation of that feature. This is done for each data point in each feature.

Next, the covariance matrix of each feature is computed. Covariance, essentially, measures how two variables change when one changes. For example, a positive covariance indicates that when one variable increases, the other variable increases. A negative covariance would indicate that one decreases when the other increases, and a covariance of 0 would indicate that they do not change in relation to each other and are unrelated. The formula for covariance is as follows.

$$Cov(x, y) = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \overline{X})(Y_i - \overline{Y})$$

This forms a matrix with the covariances, where the diagonals along the matrix are variances, and the off-diagonals are covariances. This matrix is then used to quantify the linear relationships between standardized variables.

Next, eigenvalues and eigenvectors are the core components of PCA. An eigenvector is a direction in space along which the variables change the most. Eigenvectors, essentially, are principal components and create new axes that capture the maximum variance. To mathematically define it, $Av = \lambda v$, where A is the covariance matrix, v is the eigenvector, and λ is the eigenvalue. An eigenvalue tells you exactly how much variance exists along the direction defined by the eigenvector. This means that the total variance is defined as $\sum_{i=1}^{n} \lambda_i$.

Finally, the explained variance ratio is calculated as being $Explained\ Variance\ Ratio = \frac{\lambda_i}{\sum\lambda}$. This is calculated for each individual Principal Component to find the percentage for each one. Then, the three components are added together to find the total Explained Variance.

The second metric is the mean silhouette score and the standard deviation of the silhouette scores. This is most essential since it measures how well-separated the clusters are, meaning how well of a job the algorithm did at grouping galaxies. Each point gets a silhouette value from -1 to 1, calculated as explained above. This time, to find the mean silhouette score, we take the mean of the silhouette scores of every point.

The third metric is the cluster stability. To do this, we use the Adjusted Rand Index (ARI), which, fundamentally, compares two clusters and describes how similar to each other they are, answering the question, “If the clustering algorithm were done again, would it reproduce essentially the same clusters?” ARI is generally on a scale from 0 to 1, with 0 representing the clusters being perfectly random and 1 representing perfect clustering. However, ARI can be less than 0, which would mean that the clustering was worse than random. This is uniquely distinct from the mean silhouette score because it measures the similarity between clusters, whereas the mean silhouette score represents the grouping of points within clusters. This is calculated by taking two clusters and calculating both ‘a’, representing the number of galaxy pairs in the same cluster in both clusterings, and ‘b’, representing the number of galaxy pairs in different clusters in both clusterings. Then, the Rand Index is computed as $RI = \frac{a+b}{total\ pairs}$. To find the Adjusted Rand Index, we calculate $ARI = \frac{RI - Expected\ RI}{1 - expected\ RI}$, with the expected RI being calculated inside each k-means within the algorithm, measuring the similarity expected between two random clusterings of the same size.

Results

Principal Component Analysis

After z-score standardization was applied to each feature being tested, PCA was applied to reduce the dimensionality of the features. To ensure that PCA is an accurate representation of the variables, PCA Explained Variance was calculated as described above. Through the calculations described, the results were found to be:

PC1: 55.09%
PC2: 19.13%
PC3: 14.47%

Combined, the principal components accounted for 88.67% of the total variance in the dataset used. This was done as a preliminary analysis to ensure that the three dimensions that the dataset was reduced to were accurate to continue with the methodology. A total variance coverage of 88.67% indicates that the majority of the variability present in the dataset could be accurately represented by three dimensions. This signifies that clustering done using the three-dimensional space through the three principal components will be representative of the original variables introduced.

K-value

Next, k-means clustering was applied as explained above and performed for values of k=2 through k=6 inclusive. The k-values are as seen below.

k = 2: 0.4228
k = 3: 0.3084
k = 4: 0.3610
k = 5: 0.4109
k = 6: 0.3538

As seen, the highest silhouette score was found at k=2, meaning that two clusters were found to be optimal for this experiment.

Cluster Separation

Using k = 2, the silhouette scores for each data point (galaxy) were found and analyzed statistically as described in the methodology. The statistics deemed significant for this study are as follows.

Mean Silhouette Score: 0.4228
Standard Deviation of Silhouette Scores: 0.1570

The mean silhouette score, being representative of how accurately the algorithm clustered data, indicates that clusters are separated moderately. Generally, a value less than 0.5 indicates moderate separation, with values close to 0 representing overlapping clusters. In addition, a baseline analysis was done, producing results of clusters near zero, confirming that the mean silhouette score found is statistically significant. Values greater than 0.5 generally indicate very strong clustering. A mean silhouette score of 0.4228 indicates that clustering was moderately strong and close to being considered very strong, but not strong enough to be considered near-perfect. Physically, this means that there were likely galaxies that were on the edges of clusters, meaning they likely fit within the cluster, but somewhat varied from the rest of the items in the cluster.

Cluster Stability

The Adjusted Rand Index, as described above, is a measure of cluster stability, indicating whether the clustering done by the algorithm is consistent if clustering occurs consecutively. To measure this, clustering was completed ten times consecutively. Using the calculations described in the methodology, the results are as seen below.

Mean ARI: 0.8781
ARI Standard Deviation: 0.2988

A mean ARI of 0.8781 is very strong and indicates that the algorithm performed clustering with high reproducibility across trials. ARI values near 1 represent almost-identical clusters, meaning that 0.8781 indicates the clusters were extremely similar. Statistically, this means that the clustering from different trials agrees 87.81% more than what would be expected by chance clustering, after adjusting for randomness. However, the standard deviation causes slight concern since it indicates that certain clusters were very different, which might be a limitation, although it might also be since only ten trials occurred. However, the high mean ARI likely outweighs this since it shows that the clustering is generally stable and reproducible.

Interpretation

Taking all of these metrics into account, the unsupervised clustering algorithm remains high in variance after dimensionality reduction, moderate in clustering separation, and strong in clustering stability. It is impossible to define “accuracy” for an unsupervised algorithm, since accuracy requires a label on the data to determine whether the clusters are “correct” or not. However, to define an accuracy in a single metric, it would be acceptable to simply use the Mean Silhouette Score, as this is seen as the accuracy of how well galaxies are clustered together by the algorithm. This was chosen instead of the PCA Explained Variance or the mean ARI as the metric of evaluation because explained variance is simply to ensure that the dimensionality-reduction is accurate and can be used to find our results, and the mean ARI simply ensures that our results are reproducible, but does not tell whether those reproducible

results are clustered well or not. However, a Mean Silhouette Score of 0.4228 directly indicates that the galaxies are moderately grouped. For reference, random clustering would usually produce around a mean silhouette score of 0.

Limitations

Although the results of this study are clear, there are a few limitations to the accuracy of this study and its application.

Firstly, k-means algorithms assume that clusters are spherical. K-means clustering uses Euclidean distance, but this assumes that clusters are spherical and data points are separated linearly. Although this is likely what the clusters should theoretically look like, there is a chance that the actual clustering is elongated in some other shape that is not spherical. If so, that would mean that the mean silhouette score is not an accurate measurement because it assumes that the clusters are meant to be spherical. However, spherical clusters are most likely, and the moderate mean silhouette score verifies that spherical clusters fit the data, at least somewhat, meaning that it is likely that spherical clusters are accurate.

Secondly, the use of PCA creates a new limitation, although PCA is necessary for this project. PCA is a method of dimensionality reduction, meaning it identifies variables along a third axis to evaluate. However, this also means that it assumes variables are linearly related to each other. This means that if, for example, two variables have a non-linear relationship, PCA would not be an accurate representation of the relationship between variables. However, this was minimized through the PCA explained variance, which ensures that most of the data is accurately represented in the third dimension.

Another limitation is the relatively small dataset compared to the population. Although 175 galaxies in the SPARC database are more than enough to assume normality, 175 galaxies are an extremely small portion of all existing galaxies, assuming trillions of galaxies exist. This means that although 175 is a good approximation of the population, it means that a few items may fluctuate when compared to the real population. For example, silhouette scores likely have a much higher standard deviation than calculated, and the stability calculations may be slightly inflated. However, this encourages experimentation with larger datasets. In addition, there may be selection bias amongst the galaxies contained in the SPARC database itself. These galaxies were likely chosen due to quality rotation curves and quality measurements, meaning that, although the SPARC database contains a diverse range of galaxies as explained previously, there is still likely a selection bias.

Next, there are interpretation limits of the mean silhouette score. Firstly, the calculated PCA explained variance means that about 11% of variance was discarded, which means that there likely is a small difference in the calculated silhouette score and actual if this experiment were to be done without PCA. Secondly, the silhouette score being moderate and not strong means that there is likely a high standard deviation amongst the silhouette scores of individual data points, meaning that our mean would likely change when reproduced.

Application/Reproducibility

Reproducibility

With these results in mind, this study was designed to be computationally reproducible. The full pipeline of scripts is described above in methodology, and the code for each script is shown respectively in Appendix A, Appendix B, and Appendix C.

The trained model can be applied to any dataset that contains the six elements described in the methodology. This is also a limitation, since you'll need access to a dataset with these specific features for it to be reproducible using the algorithm.

However, the algorithm itself is very reproducible. As calculated in the methodology section, the cluster stability, represented by the mean Adjusted Rand Index, was calculated to be 0.8781 on a scale maximized at 1. That is a very high statistic, which proves that the algorithm is highly reproducible.

Application

In terms of application of results, the moderate mean silhouette score and high reproducibility signify that the structure of galaxy rotation curves in the search for dark matter shows persistent organization. This may be specifically applicable to dark matter halo detection. Because the methodology groups galaxies according to their rotation curve geometry, the resulting clusters may reflect structural differences in galaxies that create the basis for dark matter, such as differences in the outer rotation curve velocities, which were a core component used in the dimensionality-reduction parameters. This is made especially interesting due to the fact that the core parameters entered into the dimensionality-reduction are specifically relevant to dark matter haloes, such as the outer baryonic fraction and the outer curve velocity, with their importance being described above in the literature review.

This study calls upon future studies to include measurements of accuracy for their methodologies and suggests increased use of unsupervised machine learning methods due to the preference discussed in the introduction and literature review, as well as their accuracy as defined by this research. In addition, future studies should consider using other methods for clustering, such as GMMs (Gaussian Mixture Models) or DBSCAN, to determine the accuracy for those in comparison to the accuracy of k-means

Importantly, this study does not claim that these identified groupings directly or linearly relate to the amount of dark matter contained in a galaxy. Instead, what this study identifies is groupings of galaxies that have structural differences. These differences make up the basis for the detection of dark matter, which may suggest that galaxies in a certain cluster have a different dark matter composition than galaxies in another. These results should be interpreted as identifying tendencies and features of galaxies and grouping them amongst those tendencies, and not as a direct correlation to dark matter.

All together, these results should be taken as showing that unsupervised machine learning algorithms can moderately accurately identify differences in galactic rotation curves that may suggest the presence of dark matter as compared to Baryonic mass predictions.

Appendix A

Prepare_data.py

```python
import os
import glob
import numpy as np
import pandas as pd

PROJECT_ROOT = os.path.abspath(os.path.join(os.path.dirname(__file__), ".."))
DATA_DIR = os.path.join(PROJECT_ROOT, "data")
ROTMOD_DIR = os.path.join(DATA_DIR, "Rotmod_LTG")
FEATURES_OUT = os.path.join(DATA_DIR, "features.csv")

# Choice of stellar mass-to-light scaling (SPARC V_disk/V_bul are for M/L=1)
MU = 0.5  # change if you want to test other M/L values

def safe_sqrt(x):
    x = np.array(x, dtype=float)
    x[x < 0] = np.nan
    return np.sqrt(x)

def parse_rotmod_dat(path):
    """
    Read a *_rotmod.dat file. These have comment lines beginning with '#'
    and then whitespace-separated numeric columns like:
    Rad  Vobs  errV  Vgas  Vdisk  Vbul  SBdisk  SBbul
    """
    try:
        df = pd.read_csv(path, comment="#", delim_whitespace=True, header=None)
    except Exception as e:
        print(f"Failed to read {path}: {e}")
        return None

    if df.shape[1] < 6:
        # not enough columns
        return None

    arr = df.values.astype(float)
    # columns: 0=R,1=Vobs,2=errV,3=Vgas,4=Vdisk,5=Vbul, (6,7 optional SB)
    R = arr[:, 0]
    Vobs = arr[:, 1]
    # errV = arr[:,2]  # not used directly in features but kept if needed
    Vgas = arr[:, 3] if arr.shape[1] > 3 else np.zeros_like(Vobs)
    Vdisk = arr[:, 4] if arr.shape[1] > 4 else np.zeros_like(Vobs)
    Vbul = arr[:, 5] if arr.shape[1] > 5 else np.zeros_like(Vobs)
```

```
# compute baryonic velocity squared and baryonic velocity
V_bary_sq = Vgas**2 + MU * (Vdisk**2 + Vbul**2)
V_bary = safe_sqrt(V_bary_sq)

# residuals: observed minus baryonic
res = Vobs - V_bary

# features (outer region defined as outer third of radii)
res_max = np.nanmax(res) if res.size > 0 else np.nan
n = len(R)
outer_start = max(0, int(n * 2 / 3))
outer_idx = np.arange(outer_start, n) if n > 0 else np.arange(0)

if len(outer_idx) == 0:
    outer_idx = np.arange(n)

res_outer_mean = np.nanmean(res[outer_idx]) if len(outer_idx) > 0 else np.nan
res_outer_median = np.nanmedian(res[outer_idx]) if len(outer_idx) > 0 else np.nan

# slope in outer region
try:
    if len(outer_idx) >= 2:
        p = np.polyfit(R[outer_idx], res[outer_idx], 1)
        res_outer_slope = float(p[0])
    else:
        res_outer_slope = np.nan
except Exception:
    res_outer_slope = np.nan

v_flat = float(np.nanmedian(Vobs[outer_idx])) if len(outer_idx) > 0 else np.nan

# baryonic fraction outer: mean(V_bary^2 / V_obs^2) over outer region (safe handling)
with np.errstate(divide="ignore", invalid="ignore"):
    frac = (V_bary[outer_idx] ** 2) / (Vobs[outer_idx] ** 2)
baryon_frac_outer = float(np.nanmean(frac)) if len(outer_idx) > 0 else np.nan

n_points = int(n)
```

```
    return {
        "res_max": float(res_max),
        "res_outer_mean": float(res_outer_mean),
        "res_outer_median": float(res_outer_median),
        "res_outer_slope": float(res_outer_slope),
        "v_flat": float(v_flat),
        "baryon_frac_outer": float(baryon_frac_outer),
        "n_points": n_points,
    }

def main():
    pattern = os.path.join(ROTMOD_DIR, "*_rotmod.dat")
    files = sorted(glob.glob(pattern))
    if len(files) == 0:
        print(f"No '*_rotmod.dat' files found in {ROTMOD_DIR}. Make sure you downloaded and unzipped Rotmod_LTG
        return

    features = []
    for fpath in files:
        feats = parse_rotmod_dat(fpath)
        if feats is None:
            print("Skipping (could not parse):", os.path.basename(fpath))
            continue
        name = os.path.basename(fpath).replace("_rotmod.dat", "")
        feats["galaxy"] = name
        features.append(feats)

    if len(features) == 0:
        print("No features extracted. Check Rotmod_LTG contents and file parsing assumptions.")
        return

    df = pd.DataFrame(features)
    df.to_csv(FEATURES_OUT, index=False)
    print("Saved features to", FEATURES_OUT)
    print(df.head())

if __name__ == "__main__":
    main()
```

Appendix B

Train_model.py

```python
import os
import joblib
import numpy as np
import pandas as pd
from sklearn.preprocessing import StandardScaler
from sklearn.decomposition import PCA
from sklearn.cluster import KMeans, DBSCAN
from sklearn.metrics import silhouette_score

PROJECT_ROOT = os.path.abspath(os.path.join(os.path.dirname(__file__), ".."))
DATA_DIR = os.path.join(PROJECT_ROOT, "data")
MODELS_DIR = os.path.join(PROJECT_ROOT, "models")
RESULTS_DIR = os.path.join(PROJECT_ROOT, "results")
os.makedirs(MODELS_DIR, exist_ok=True)
os.makedirs(RESULTS_DIR, exist_ok=True)

FEATURES_IN = os.path.join(DATA_DIR, "features.csv")

def load_features(path):
    df = pd.read_csv(path)
    df = df.dropna(subset=['res_outer_mean','res_max'])  # keep rows with these
    df = df.reset_index(drop=True)
    return df

from sklearn.metrics import silhouette_score, silhouette_samples, adjusted_rand_score
from sklearn.utils import resample
from scipy.stats import f_oneway, kruskal

def main():
    df = load_features(FEATURES_IN)
    galaxy_names = df['galaxy'].values
    X = df[['res_max','res_outer_mean','res_outer_slope','v_flat','baryon_frac_outer','n_points']].values

    # -------- Scaling --------
    scaler = StandardScaler().fit(X)
    Xs = scaler.transform(X)
    joblib.dump(scaler, os.path.join(MODELS_DIR, "scaler.joblib"))

    # -------- PCA --------
    pca = PCA(n_components=3, random_state=0).fit(Xs)
    Xp = pca.transform(Xs)
    joblib.dump(pca, os.path.join(MODELS_DIR, "pca.joblib"))
    print("PCA explained variance ratios:", pca.explained_variance_ratio_)
```

```python
# -------- Choose Best k --------
best_k = None
best_score = -1
for k in range(2,7):
    kmeans_temp = KMeans(n_clusters=k, random_state=0).fit(Xp)
    labels_temp = kmeans_temp.labels_
    try:
        score = silhouette_score(Xp, labels_temp)
    except Exception:
        score = -1
    print("k =", k, "silhouette =", score)
    if score > best_score:
        best_score = score
        best_k = k

print("Best k chosen:", best_k)

# -------- Final Clustering --------
kmeans = KMeans(n_clusters=best_k, random_state=0).fit(Xp)
labels = kmeans.labels_
joblib.dump(kmeans, os.path.join(MODELS_DIR, "kmeans.joblib"))

# -------- 1️⃣ Silhouette (FINAL MODEL) --------
mean_sil = silhouette_score(Xp, labels)
sil_samples = silhouette_samples(Xp, labels)
print("Final Mean Silhouette Score:", mean_sil)
print("Silhouette Std Dev:", sil_samples.std())

# -------- 2️⃣ Stability Test (ARI) --------
ari_scores = []
for i in range(10):
    kmeans_new = KMeans(n_clusters=best_k, random_state=i).fit(Xp)
    labels_new = kmeans_new.labels_
    ari = adjusted_rand_score(labels, labels_new)
    ari_scores.append(ari)

print("Mean ARI (stability):", np.mean(ari_scores))
print("ARI Std Dev:", np.std(ari_scores))
```

```
    # -------- Save Results --------
    df_out = df.copy()
    df_out['cluster'] = labels
    df_out.to_csv(os.path.join(RESULTS_DIR, "clustered.csv"), index=False)
    print("Saved clustered results to results/clustered.csv")

    # -------- 3️⃣ Physical Validation (ANOVA + Kruskal) --------
    variable = "baryon_frac_outer"
    groups = [group[variable].values for name, group in df_out.groupby("cluster")]

    f_stat, p_val = f_oneway(*groups)
    print("ANOVA F:", f_stat)
    print("ANOVA p-value:", p_val)

    h_stat, p_kw = kruskal(*groups)
    print("Kruskal H:", h_stat)
    print("Kruskal p-value:", p_kw)



if __name__ == "__main__":
    main()
```

Appendix C

Predict_new.py

```python
import os, sys, argparse, joblib, pandas as pd, numpy as np

PROJECT_ROOT = os.path.abspath(os.path.join(os.path.dirname(__file__), ".."))
MODELS_DIR = os.path.join(PROJECT_ROOT, "models")
RESULTS_DIR = os.path.join(PROJECT_ROOT, "results")
os.makedirs(RESULTS_DIR, exist_ok=True)

def load_models():
    for name in ("scaler.joblib","pca.joblib","kmeans.joblib"):
        p = os.path.join(MODELS_DIR, name)
        if not os.path.exists(p):
            raise FileNotFoundError(f"Missing model file: {p}")
    scaler = joblib.load(os.path.join(MODELS_DIR,"scaler.joblib"))
    pca = joblib.load(os.path.join(MODELS_DIR,"pca.joblib"))
    kmeans = joblib.load(os.path.join(MODELS_DIR,"kmeans.joblib"))
    return scaler, pca, kmeans

def predict_df(new_df, scaler, pca, kmeans, train_cluster_df=None):
    needed = ['res_max','res_outer_mean','res_outer_slope','v_flat','baryon_frac_outer','n_points']
    for c in needed:
        if c not in new_df.columns:
            raise ValueError(f"Input missing column: {c}")
    X = new_df[needed].astype(float).values
    Xs = scaler.transform(X)
    Xp = pca.transform(Xs)
    labels = kmeans.predict(Xp)
    out = new_df.copy()
    out['cluster'] = labels
    # build simple textual labels from training medians if given
    if train_cluster_df is not None:
        med = train_cluster_df.groupby('cluster')['res_outer_mean'].median().sort_values(ascending=False)
        mapping = {}
        rank_label = {0:"high-DM-like",1:"mid-DM-like",2:"low-DM-like"}
        for i, c in enumerate(med.index):
            mapping[c] = rank_label.get(i, f"cluster-{i}")
        out['label'] = out['cluster'].map(mapping).fillna("cluster-unknown")
    else:
        out['label'] = out['cluster'].apply(lambda x: f"cluster-{x}")
    return out
```

```python
def main():
    parser = argparse.ArgumentParser()
    parser.add_argument("--csv", default=None, help="CSV input file")
    args = parser.parse_args()

    try:
        scaler, pca, kmeans = load_models()
    except Exception as e:
        print("ERROR loading models:", e)
        sys.exit(1)
    print("Models loaded.")

    train_cluster_df = None
    clustered_path = os.path.join(PROJECT_ROOT, "results", "clustered.csv")
    if os.path.exists(clustered_path):
        try:
            train_cluster_df = pd.read_csv(clustered_path)
            print("Loaded training clustered.csv for label mapping.")
        except Exception:
            train_cluster_df = None

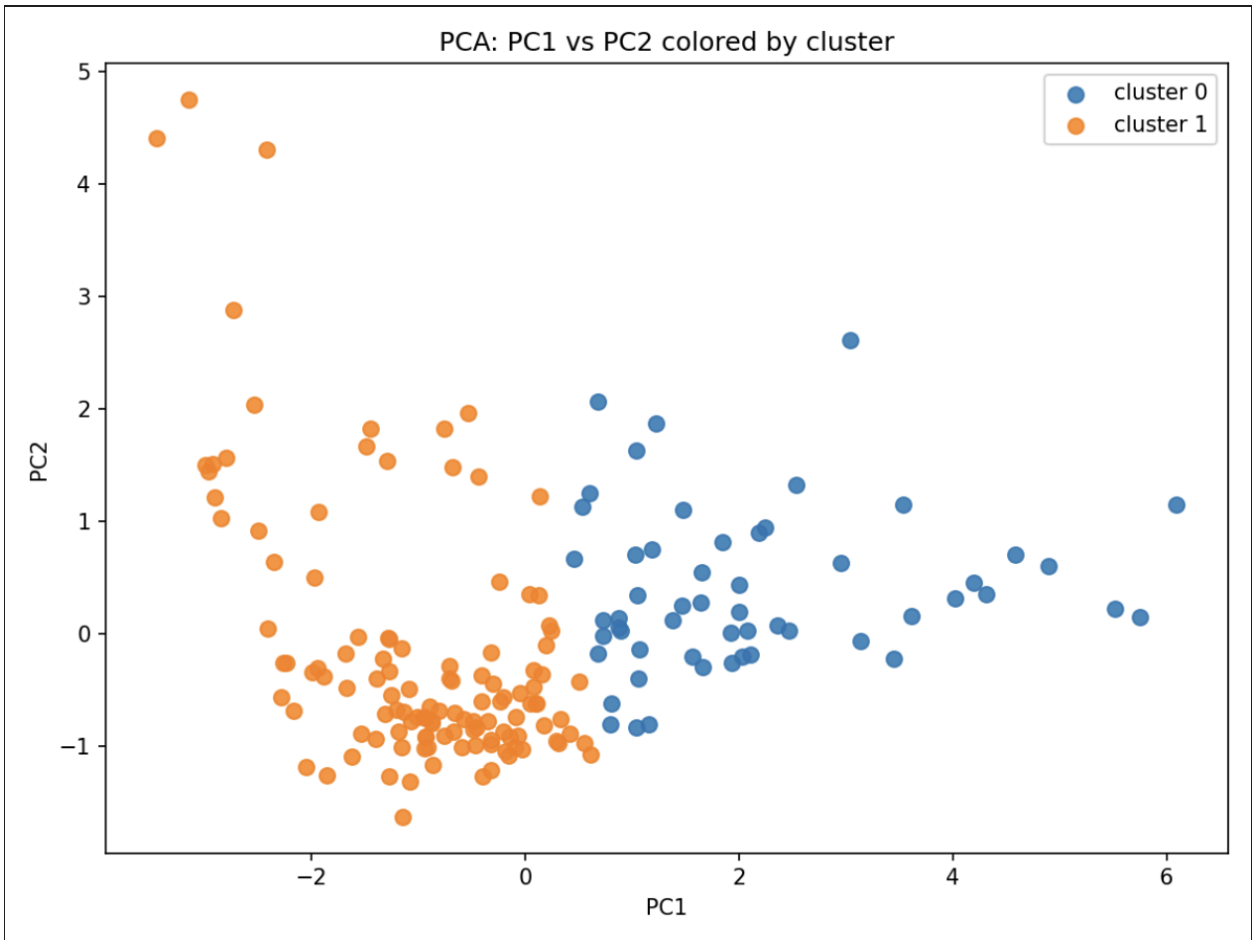
    if args.csv:
        if not os.path.exists(args.csv):
            print("ERROR: input CSV not found:", args.csv)
            sys.exit(1)
        new_df = pd.read_csv(args.csv)
        print("Loaded", args.csv, "rows:", len(new_df))
    else:
        # interactive
        print("Enter comma-separated values for: res_max,res_outer_mean,res_outer_slope,v_flat,baryon_frac_outer,
        s = input("values: ").strip()
        vals = [float(x.strip()) for x in s.split(",")]
        cols = ['res_max','res_outer_mean','res_outer_slope','v_flat','baryon_frac_outer','n_points']
        new_df = pd.DataFrame([dict(zip(cols, vals))])
        print("Using interactive input row.")

    try:
        out_df = predict_df(new_df, scaler, pca, kmeans, train_cluster_df)
    except Exception as e:
        print("ERROR during prediction:", e)
        sys.exit(1)
```

```python
    out_path = os.path.join(RESULTS_DIR, "prediction_new.csv")
    out_df.to_csv(out_path, index=False)
    print("Saved predictions to", out_path)
    print(out_df.to_string(index=False))

if __name__ == "__main__":
    main()
```

Appendix D


PCA: PC1 vs PC2 colored by cluster
cluster 0
cluster 1
PC2
PC1
5
4
3
2
1
0
−1
−2
0
2
4
6

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